

Basab : Retail_Analytics_Project:

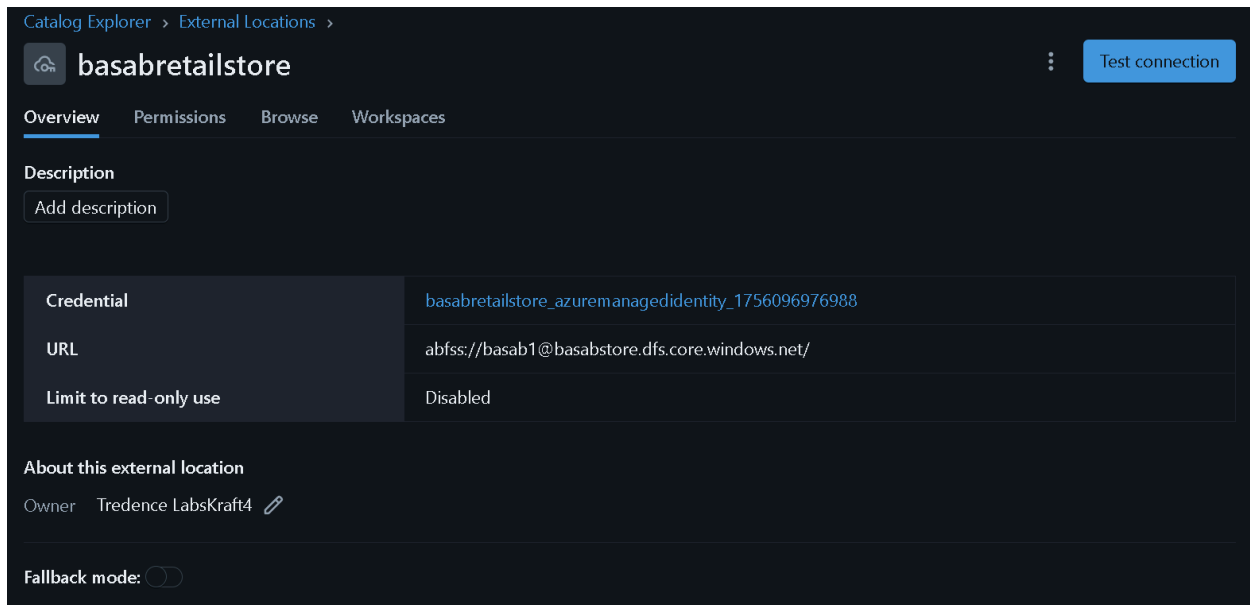
Email :

Tredence LabsKraft4

tredence4@michaelkloudkrafthotmail.onmicrosoft.com

Unity Catalog Creation :-

External Location Created for Unity Catalog



The screenshot shows the 'basabretailstore' external location configuration in the Unity Catalog interface. The breadcrumb path is 'Catalog Explorer > External Locations > basabretailstore'. There are tabs for 'Overview', 'Permissions', 'Browse', and 'Workspaces', with 'Overview' selected. A 'Test connection' button is in the top right. The 'Description' section has an 'Add description' button. A table lists configuration details: Credential (basabretailstore_azuremanagedidentity_1756096976988), URL (abfss://basab1@basabstore.dfs.core.windows.net/), and Limit to read-only use (Disabled). The 'About this external location' section shows the owner as 'Tredence LabsKraft4' with an edit icon. At the bottom, there is a 'Fallback mode' toggle switch.

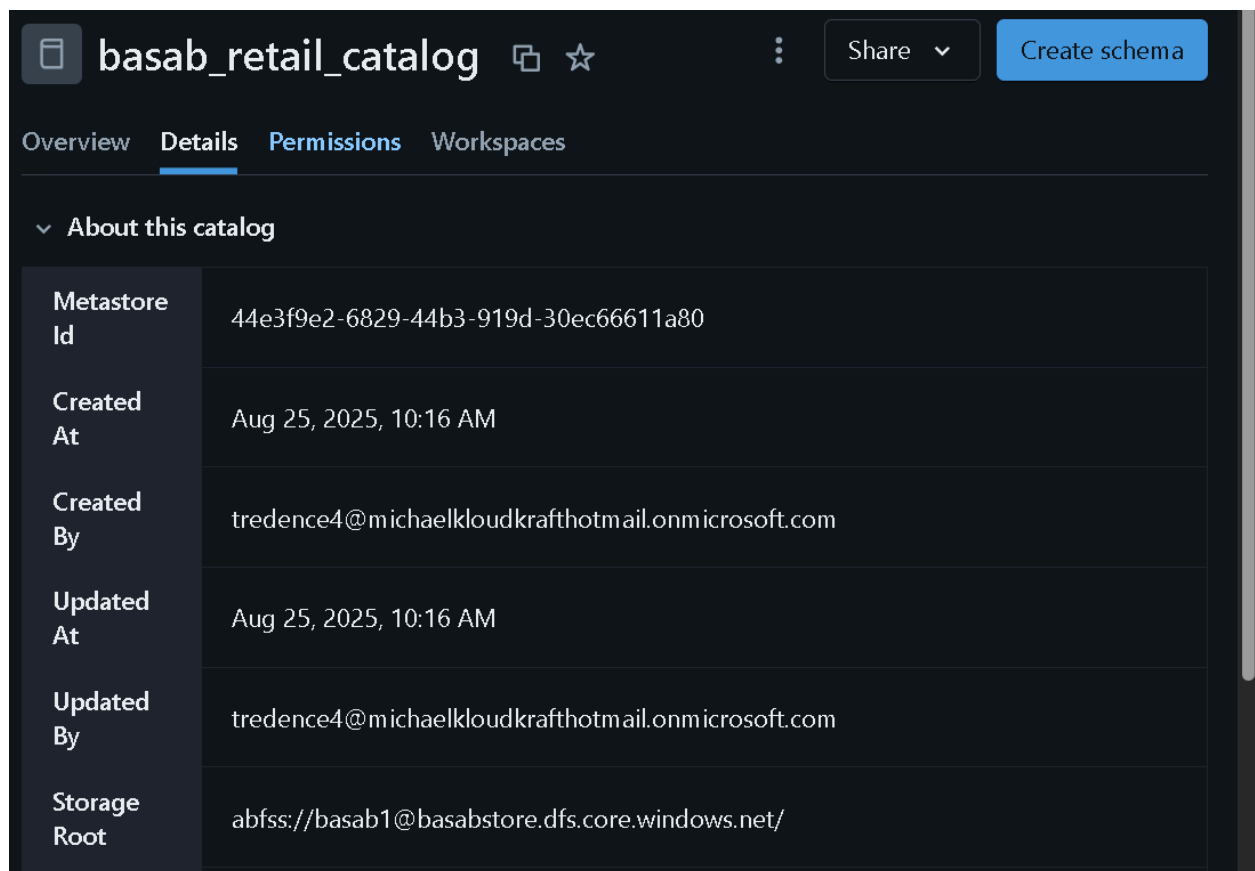
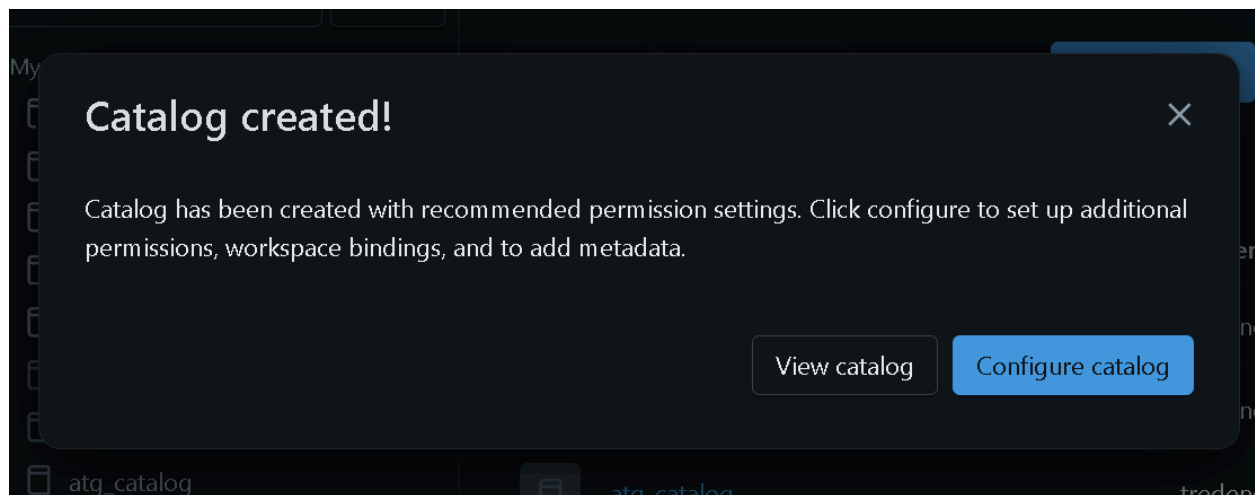
basabretailstore	
Credential	basabretailstore_azuremanagedidentity_1756096976988
URL	abfss://basab1@basabstore.dfs.core.windows.net/
Limit to read-only use	Disabled

About this external location

Owner: Tredence LabsKraft4

Fallback mode: ☐

Unity Catalog Created with External ADLS Location:



In my container :

Home > basabstore | Containers >

basab1 ...
Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

+ Add Directory | ↑ Upload | ↻ Refresh | 🗑 Delete | 📄 Copy | 📄 Paste | 🏷 Rename | 🔗 Acquire lease

basab1

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive)

Showing all 3 items

☐	Name	Last modified	Access t
☐	__unitystorage	8/25/2025, 10:49:00 AM	
☐	checkpoints	8/25/2025, 10:49:12 AM	
☐	incoming_raw	8/25/2025, 10:26:37 AM	

Ingesting Data into bronze :

Workspace

Catalog

Type to search...

For you All

- My organization
 - tred_d1
 - system
 - basab_catalog2
 - basab_retail_catalog
 - basab_iot_schema
 - bronze_orders
 - default
 - information_schema
 - basab_unity_catalog
 - aks_catalog
 - aks_catalog_new
 - aks_proj1
 - atg_catalog

Ingest

Python Tabs: ON Last edit was now

10:37 AM (2s) 2 Python

```
df_bronze=spark.readStream.format("csv").option("header","true").schema
(bronze_schema).load(raw_path)
```

df_bronze: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 12 more fields]

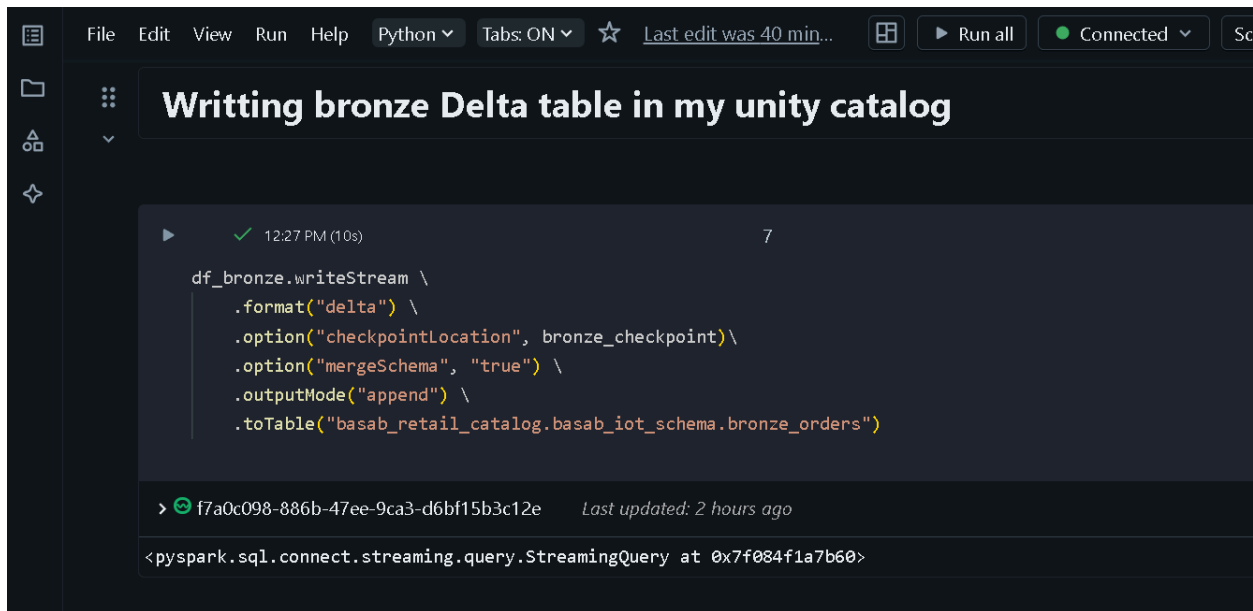
10:44 AM (2s) 3

```
%sql
use catalog basab_retail_catalog;
create schema basab_iot_schema;
```

10:44 AM (1s) 4

```
%sql
USE basab_retail_catalog.basab_iot_schema;
```

Writting Bronze Delta table in my Unity Catalog :



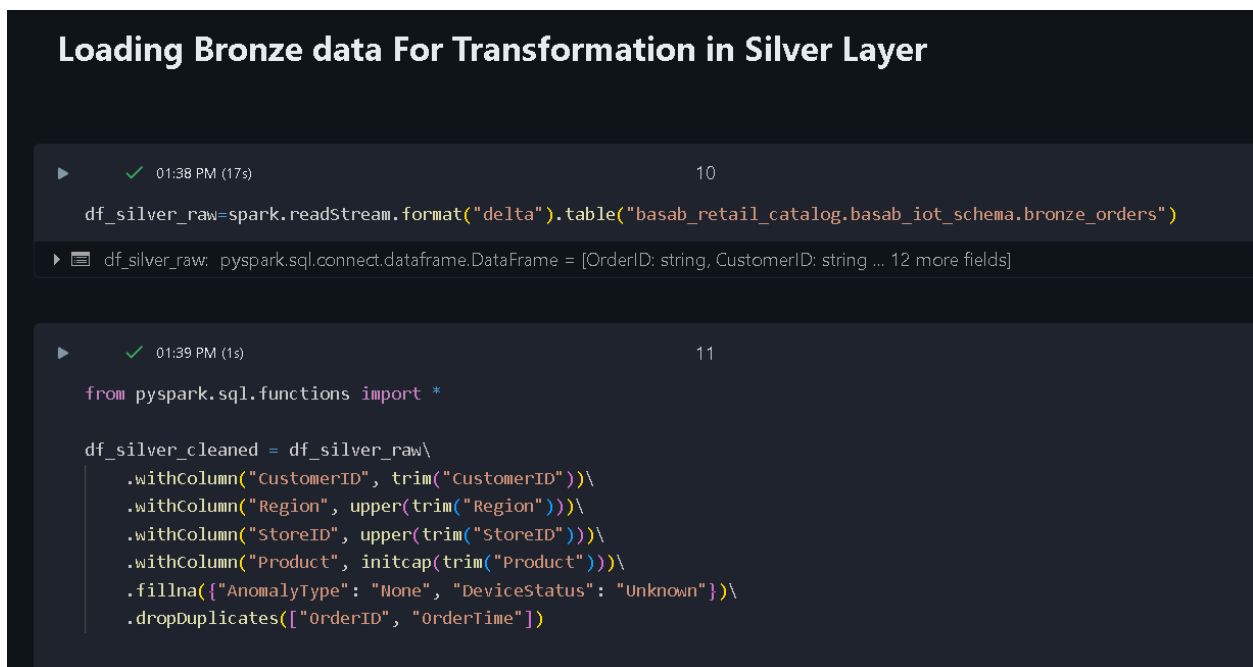
```
df_bronze.writeStream \
    .format("delta") \
    .option("checkpointLocation", bronze_checkpoint)\
    .option("mergeSchema", "true") \
    .outputMode("append") \
    .toTable("basab_retail_catalog.basab_iot_schema.bronze_orders")
```

> f7a0c098-886b-47ee-9ca3-d6bf15b3c12e Last updated: 2 hours ago

<pyspark.sql.connect.streaming.query.StreamingQuery at 0x7f084f1a7b60>

Loading Bronze Delta table for Silver layer transformation

Col names are trimmed and null values in Anomaly type and devieStatus are filled and duplicates in OrderID and OrderTime is dropped :-



```
df_silver_raw=spark.readStream.format("delta").table("basab_retail_catalog.basab_iot_schema.bronze_orders")
```

df_silver_raw: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 12 more fields]

```
from pyspark.sql.functions import *

df_silver_cleaned = df_silver_raw\
    .withColumn("CustomerID", trim("CustomerID"))\
    .withColumn("Region", upper(trim("Region")))\
    .withColumn("StoreID", upper(trim("StoreID")))\
    .withColumn("Product", initcap(trim("Product")))\
    .fillna({"AnomalyType": "None", "DeviceStatus": "Unknown"})\
    .dropDuplicates(["OrderID", "OrderTime"])
```

Adding Business Logic for Silver Layer

Adding Anomaly FLAG

Adding NormalisedProduct Col

```
12:46 PM (< 1s) 17

silver_business = (
    df_silver_cleaned
    # Add anomaly flag
    .withColumn("HasAnomaly", when(col("AnomalyFlag") == 1, lit(True)).otherwise(lit(False)))

    # Normalize product categories
    .withColumn(
        "NormalizedProduct",
        when(col("Product").isin("Mobile", "Smartphone"), "Mobile")
        .when(col("Product").isin("Laptop", "Notebook"), "Laptop")
        .when(col("Product").isin("Smartwatch", "Watch"), "Smartwatch")
        .when(col("Product").isin("Headphones", "Earbuds", "Earphones"), "Headphones")
        .when(col("Product").isin("Tablet", "iPad"), "Tablet")
        .when(col("Product").isin("Printer"), "Printer")
        .when(col("Product").isin("Camera"), "Camera")
        .otherwise(col("Product")) # fallback → keep original
    )
)

silver_business: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 14 more fields]
```

Dropping Rows with nulls in critical Cols like : OrderID , Quantity

```
12:46 PM (< 1s) 19

# List of string columns you want to clean
string_cols = ["OrderID", "CustomerID", "StoreID", "Product"]

for c in string_cols:
    silver_business = silver_business.withColumn(c, when(trim(col(c)) == "", None).otherwise(trim(col(c))))

silver_business: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 14 more fields]
```

Dropping the Rows which has nulls in Critical Cols like :OrderID , StoreID

```
12:46 PM (< 1s) 21

critical_cols = ["OrderID", "CustomerID", "StoreID", "Product", "Quantity", "UnitPrice", "SalesAmount", "OrderTime"]

silver_business_cleaned = silver_business.dropna(subset=critical_cols)

silver_business_cleaned: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 14 more fields]
```

Writting Silver Delta table in my unity Catalog's schema :

Writting Silver Delta Table in my Unity Catalog

```
12:27 PM (5s) 25
silver_business_cleaned.writeStream \
  .format("delta") \
  .option("checkpointLocation", silver_checkpoint)\
  .option("mergeSchema", "true") \
  .outputMode("append") \
  .toTable("basab_retail_catalog.basab_iot_schema.silver_table")
```

Stream initializing...

<pyspark.sql.connect.streaming.query.StreamingQuery at 0x7f082dd86fc0>

Before Gold Layer Aggregation

Optimise Silver Delta table and Making Silver Partioned Delta table

```
12:52 PM (5m) 28
%sql
-- Partition by Region and Order Date
CREATE OR REPLACE TABLE basab_retail_catalog.basab_iot_schema.silver_orders_partitioned
USING DELTA
PARTITIONED BY (Region, OrderDate)
AS
SELECT *, CAST(OrderTime AS DATE) AS OrderDate
FROM basab_retail_catalog.basab_iot_schema.silver_table;
```

(1) Spark Jobs

```
%sql OPTIMIZE basab_retail_catalog.basab_iot_schema.silver_orders_partitioned
ZORDER BY (Product, StoreID);
```

(100) Spark Jobs

Loading Optimised Silver partitioned delta table for Gold Layer Creation (Agg. Tables)

```
02:08 PM (15s) 30
from pyspark.sql.functions import *
df_gold_raw = spark.read.table("basab_retail_catalog.basab_iot_schema.silver_orders_partitioned")

# Add OrderDate column if not already
df_gold_raw = df_gold_raw.withColumn("OrderDate", to_date(col("OrderTime")))

df_gold_raw: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 15 more fields]
```

Making all the gold aggregated tables :

```
# Daily sales
daily_sales = (
    df_gold_raw.groupBy("Region", "OrderDate")
    .agg(sum("SalesAmount").alias("TotalSales"),
        count("OrderID").alias("TotalOrders"))
)

monthly_sales = (
    df_gold_raw.withColumn("YearMonth", month(col("OrderDate")).cast("string"))
    .groupBy("Region", "YearMonth")
    .agg(sum("SalesAmount").alias("MonthlySales"),
        count("OrderID").alias("MonthlyOrders"))
)

df_gold_raw = df_gold_raw.withColumn(
    "NormalizedStoreID",
    regexp_replace("StoreID", "Store-", "")
)

device_anomaly_trend = (
    df_gold_raw.groupBy("NormalizedStoreID", "OrderDate")
    .agg(
        count(when(col("HasAnomaly") == True, 1)).alias("AnomalyCount"),
```

```

        count("OrderID").alias("TotalOrders")
    )
    .withColumn("AnomalyRate", col("AnomalyCount") / col("TotalOrders"))
)

conversion_impact = (
    df_gold_raw.groupBy("NormalizedStoreID")
    .agg(
        count(when(col("HasAnomaly") == True, 1)).alias("FailedDeviceOrders"),
        count("OrderID").alias("TotalOrders")
    )
    .withColumn("ConversionImpact", 1 - (col("FailedDeviceOrders") / col("TotalOrders")))
)

store_tiers = (
    df_gold_raw.groupBy("NormalizedStoreID")
    .agg(sum("SalesAmount").alias("TotalSales"))
    .withColumn(
        "StoreTier",
        when(col("TotalSales") > 50000, "High")
        .when(col("TotalSales") > 20000, "Medium")
        .otherwise("Low")
    )
)

weighted_avg_sales = (
    df_gold_raw.groupBy("NormalizedStoreID")
    .agg(
        (sum(col("SalesAmount") * col("HasAnomaly").cast("int")) / sum(col("SalesAmount"))).alias("WeightedAnomalyImpact")
    )
)

```

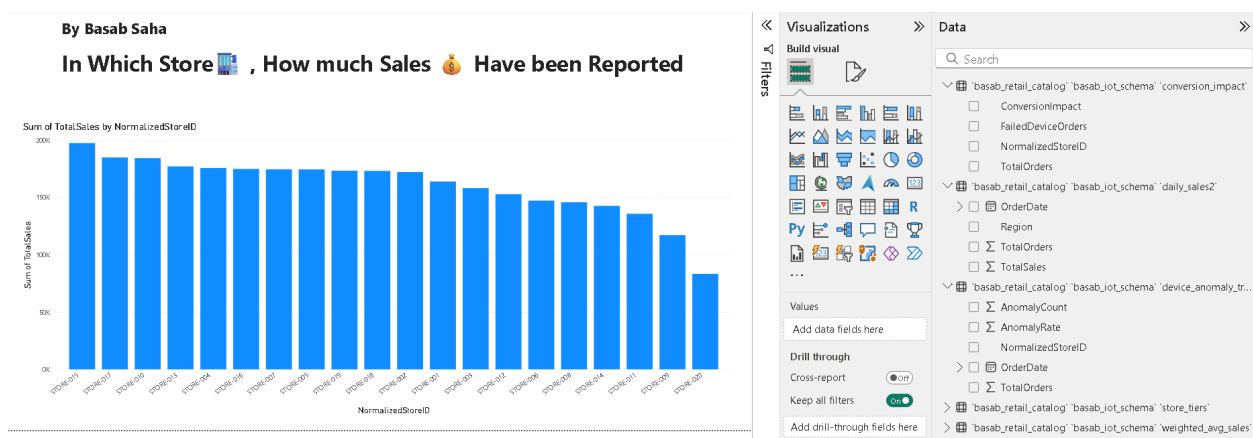

Writting aggregated gold delta tables in my unity catalog ;

```
# Daily Sales
daily_sales.write.format("delta") \
    .mode("overwrite") \
    .option("mergeSchema", "true") \
    .saveAsTable("basab_retail_catalog.basab_iot_schema.daily_sales2")

# Device Anomaly Trend
device_anomaly_trend.write.format("delta") \
    .mode("overwrite") \
    .option("mergeSchema", "true") \
    .saveAsTable("basab_retail_catalog.basab_iot_schema.device_anomaly_trend")

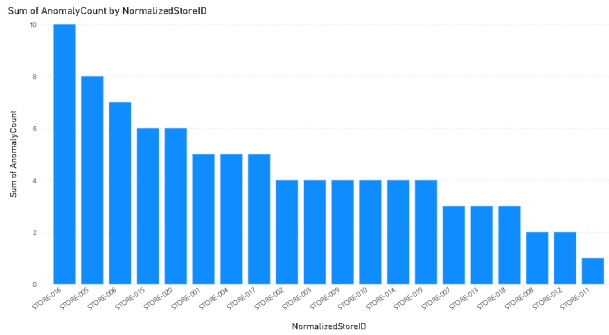
# Conversion Impact
conversion_impact.write.format("delta") \
    .mode("overwrite") \
```

PoweBI Live Dashboard 💕:



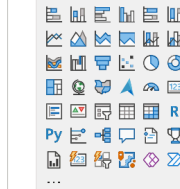
By Basab Saha

In Which Store 🏪 , How many Anomalies ✖ Have been



Visualizations

Build visual



Values

Add data fields here

Drill through

Cross-report

Keep all filters

Add drill-through fields here

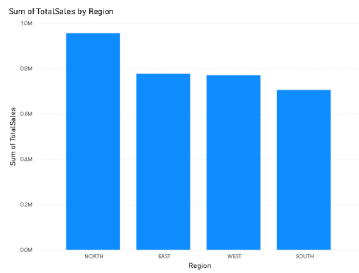
Data

Search

- basab_retail_catalog "basab_jot_schema" "conversion_impact"
 - ConversionImpact
 - FailedDeviceOrders
 - NormalizedStoreID
 - TotalOrders
- basab_retail_catalog "basab_jot_schema" "daily_sales2"
 - OrderDate
 - Region
 - TotalOrders
 - TotalSales
- basab_retail_catalog "basab_jot_schema" "device_anomaly_tr..."
 - AnomalyCount
 - AnomalyRate
 - NormalizedStoreID
 - OrderDate
 - TotalOrders
- basab_retail_catalog "basab_jot_schema" "store_tiers"
- basab_retail_catalog "basab_jot_schema" "weighted_avg_sales"

By Basab Saha

In Which Region 🌐 , How much Sales 💰 Have been Reported



By Basab Saha

In Which Store 🏪 , How much Anomaly chances % Reported

